

On the lattice one flow step has a uniformly bounded truncation error: the obstruction is decimation

The lattice flow is independent of the coupling, because the g^2 in Lüscher's equation (1.4) cancels the $1/g^2$ of the Wilson action, and one doubling step runs it for a dimensionless time $t/a^2 = 1/2$. Its linearization has coefficients that are derivatives of a fixed smooth function on the compact manifold $G^\mathcal{E}$, hence bounded by pure numbers, and the plaquette angles are bounded by the compactness of the group. So the Jacobian bound of the Jacobian note,

$$|D\Phi_t(x, y)| \leq e^{2t\|G\|_\infty} K_t^{\text{free}}(x - y), \quad 2t\|G\|_\infty \leq 2 \cdot \frac{a^2}{2} \cdot \frac{\pi}{a^2} = \pi,$$

gives a **pure number** for a doubling step, uniformly in the coupling, the scale, the volume and the gauge group up to its Casimir. Truncating the conjugated Hamiltonian at a range of K lattice spacings therefore costs Ce^{-cK} with C, c pure numbers, and K of order three suffices. This corrects the typical-field note and the Jacobian note, which located the difficulty in the supremum of the field strength: that supremum is unbounded only in the continuum at fixed physical scale, which is where the Euclidean constructive programme meets it, and it is bounded within one lattice renormalization step. What remains once truncation is free is the step that actually coarsens the theory: **decimation**, the removal of degrees of freedom, which the blocking-criterion note shows loses when performed by projection, together with the proliferation of couplings that any decimation produces. Constants explicit; nothing promoted.

1. The lattice flow carries no coupling

With the Wilson action $S_w(U) = \frac{1}{g^2} \sum_p \text{Re tr}\{1 - U_p\}$ and the flow

$$\dot{V}_s(x, \mu) = -g^2 \{\partial_{x, \mu} S_w(V_s)\} V_s(x, \mu)$$

(Lüscher, arXiv:1006.4518v3, equations (1.3)–(1.4); passage level via the local companion), the factor g^2 cancels the $1/g^2$ in S_w , so

$$\dot{V}_s(x, \mu) = -\{\partial_{x, \mu} \sum_p \text{Re tr}(1 - V_{s,p})\} V_s(x, \mu)$$

depends on no coupling. The right-hand side is a fixed smooth vector field on the compact manifold $G^\mathcal{E}$, and the same holds for its derivatives.

Consequence. The flow map Φ_t , its Jacobian $D\Phi_t$ and every bound on them are functions of the dimensionless flow time t/a^2 and of the gauge group alone.

2. One doubling step is a flow time of one half

Taking the smearing radius $\sqrt{8t}$ equal to the new lattice spacing $2a$ gives $t = a^2/2$, that is $t/a^2 = 1/2$. Two bounds then apply.

Curvature. The plaquette variable lies in the compact group, so the plaquette angle satisfies $|a^2 G| \leq \pi$ in the fundamental normalization and $\|G\|_\infty \leq \pi/a^2$; the adjoint action in the linearized flow of the Jacobian note Proposition 1 multiplies this by a Casimir factor c_A of order one. Hence

$$2t\|G\|_\infty \leq 2 \cdot \frac{a^2}{2} \cdot \frac{c_A \pi}{a^2} = c_A \pi,$$

a pure number.

Coefficients. The linearization of the lattice flow is a discrete parabolic equation whose coefficients are second derivatives of a fixed smooth function on a compact manifold, so they are bounded by pure numbers; run for dimensionless time $1/2$, its kernel obeys

$$|D\Phi_t(x, y)| \leq C_1 e^{-c_1|x-y|/a},$$

the standard exponential bound for a discrete parabolic equation with bounded coefficients over a time of order the diffusive one, with C_1, c_1 depending only on G and on the lattice geometry.

3. The truncation error

Proposition. Truncating the conjugated Hamiltonian of the flow-conjugation note to range Ka costs a relative error

$$\varepsilon(K) \leq C e^{-cK},$$

with C, c pure numbers depending only on the gauge group and the lattice geometry, uniformly in the coupling g , the lattice spacing a , the volume and the flow time of one doubling. In particular $K = 3$ gives an error of order e^{-3c} , and the effective Hamiltonian after one step has range one and a half new lattice spacings.

Proof. Section 2 bounds the Jacobian kernel by $C_1 e^{-c_1|x-y|/a}$ uniformly; the discarded part of the conjugated kinetic operator is the sum of its matrix elements at separation beyond Ka , which is bounded by the geometric tail of that estimate. The magnetic term after conjugation is $V \circ \Phi_t$, a function of the flowed links, whose dependence on a link beyond range Ka is bounded by the same kernel. $\square$

4. Correction to the two previous notes

The Jacobian note derived the condition $t\|G\|_\infty \lesssim 1$ and read it as a small-field condition, and the typical-field note concluded that the obstruction is the supremum of the field strength over the volume, hence probabilistic and Euclidean. Both statements hold in the continuum at fixed physical scale, where $\|G\|_\infty$ is unbounded and the Gaussian tail is the right tool. Within one lattice step they do not apply: the compactness of the gauge group bounds $a^2\|G\|_\infty$ by π , and the flow time of a doubling is $a^2/2$, so the product is a pure number and no configuration is large-field **relative to the current spacing**.

The large-field problem of the constructive programme is therefore a statement about many steps rather than one: a configuration that is harmless at its own scale can be large relative to a much finer spacing, and it is the accumulation over the $n \simeq (2b_0 \log 2)^{-1} g_{UV}^{-2}$ steps that produces it. The corrected reading is that the difficulty lives in the composition of the steps and in what each step does to the form of the Hamiltonian, and the tail estimate belongs to the former.

5. What is left once truncation is free

Conjugation is exact and truncation is cheap, and neither reduces the number of degrees of freedom: after both, the theory still lives on $L^2(G^\mathcal{E})$ with the original lattice. A renormalization step must also **decimate**, replacing the fine links by coarse ones. The notes already contain what happens then:

- decimation by projection onto block low-energy subspaces loses, because the boundary grows like M^2 while the block gap does not (blocking criterion);
- decimation by any map at all produces an effective Hamiltonian outside the Kogut–Susskind family, and carrying a controlled family through the steps is the constructive problem (blocking note §5).

So the sequence is: conjugate, which is free; truncate, which is cheap; decimate, which is the problem. The gain the flow provides is that after conjugation the magnetic term is the flowed action, smaller by the smoothing, so the decimation acts on a smoother field, and the question is whether that makes the projection estimate better than the $3M^2/8$ of the unflowed case. That is a well-posed question and it is the next one.

6. Consequence for STATE

The truncation error is settled: uniformly bounded per lattice doubling by compactness of the gauge group and the coupling-independence of the flow, with a range of about three lattice

spacings. The obstruction is decimation, and the remaining question native to this programme is whether flowing before decimating improves the projection estimate, since after conjugation the field entering the block boundary is the flowed one. The previous two notes stand as statements about the continuum at fixed physical scale, and carry the correction.